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Intercostal cryonerve block versus elastomeric infusion pump for postoperative analgesia following surgical stabilization of traumatic rib fractures

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ABSTRACT

Objective: Patients with blunt thoracic trauma requiring surgical stabilization of rib fractures (SSRF) frequently experience severe pain. Further, a rising prevalence of opioid-tolerant patients sustain traumatic injuries. The optimal pain management adjunct for concurrent use with SSRF remains uncertain. This study compared outcomes in patients undergoing SSRF with concomitant cryonerve block (CryoNB) or ropivacaine 0.2% elastomeric infusion pump (EIP).

Methods: A single-center retrospective comparative analysis was performed at a level II trauma center. A query of our institution's trauma registry of consecutive patients undergoing SSRF from October 2017 to November 2020 with either intercostal CryoNB or ropivacaine 0.2% EIP was conducted. Opioid consumption in oral morphine equivalents (OME), patient-reported pain scores by numerical rating scale, and pulmonary function measured by incentive spirometry effort (mL) were collected at baseline and on postoperative days 1–3. Results were analyzed using a linear-mixed-effects model. Length of stay (LOS), complications, and hospital charges were assessed as secondary outcomes.

Results: Twenty-six patients meeting inclusion criteria were evaluated. Patient demographics, injury, and surgical variables were similar between groups. The estimated effect for patients treated with CryoNB ($n = 14$) compared to EIP ($n = 12$) demonstrated a 25% (estimated -1.37 OME, 95% CI, -2.411 to -0.335, $p = 0.01$) reduction in hospital opioid requirements, fewer discharge opioids (41.3 mg (37.5–45) versus 175 mg (150- 200), $p = 0.03$), 22% (estimated -1.506, 95% CI, -2.722 to -0.290, $p = 0.02$) reduction in pain scores, and shorter postoperative LOS (4 days (4–5) versus 6 days (5–9.5), $p = 0.04$). Pulmonary function (estimated -48.8 mL, 95% CI, -312.74 to 215.05, $p = 0.71$), total hospital costs (CryoNB: \$90,224 ± 34,633; EIP: \$131,498 ± 73,072, $p = 0.07$), and complications were no different between cohorts.

Conclusion: The addition of intercostal CryoNB as an adjunct to multimodal pain management in trauma patients undergoing surgical fixation of rib fractures may be of benefit. Based on our early data, this technique appears to be promising in reducing opioid requirements and providing an extended duration of pain control without increased costs or complications.

Introduction

Traumatic rib fractures are painful and disabling, comprising 9% of all hospital admissions per year [1,2]. Failure to achieve adequate pain control increases pulmonary-related morbidity, risk of developing chronic pain, and increased hospital costs [3]. Treatment of traumatic rib fracture pain is frequently managed with opioids and every additional rib fracture increases opioid requirements by 11.2% [4]. Although effective for the reduction of acute and chronic pain, they carry risks beyond dependency and worsening addiction including delirium, constipation, and sedation [5,6]. Furthermore, trauma patients experience higher rates of preinjury opioid use compared to the general population

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with rates ranging between 16% and 20% [7]. The opioid epidemic in North America highlights the necessity for effective non-opioid alternatives in providing acute and post-acute pain control.

Surgical stabilization of rib fractures (SSRF) is increasingly being performed to improve acute pain and associated morbidity [8]. While SSRF has been shown to decrease postoperative pain and opioid requirements [9], high levels of pain are still reported [3]. Several adjuncts are used to augment the benefits of SSFR, namely neuraxial and local regional techniques. However, these options may be limited based on the availability of interventional specialists, feasibility of interrupting thromboembolic chemoprophylaxis, and display varying benefits due to the requirement for technical competency and limited duration of efficacy. Data are lacking regarding the ideal adjunct with SSRF for achieving optimal reduction of pain and opioid requirements.

Two existing modalities, elastomeric infusion pumps (EIP) and intercostal cryo-nerve block (CryoNB), have recently reemerged with increased adoption for the concomitant use with SSRF [8]. EIP is a continuous intercostal nerve block consisting of a catheter attached to a reservoir containing 550 mL of 0.2% ropivacaine and placed in the extrathoracic paraspinal space. The medication is delivered over a broad area at a controlled rate of 5 mL/hour. The benefits of this adjunct include ease of placement and the ability to provide continuous infusion for several days after surgery. Published studies have demonstrated an equivalent or superior reduction in pain [10] and opioid use [11] following thoracotomy compared to epidurals or following thoracoscopic approach with intravenous opioids [12,13]. EIP has been reported to reduce pain, length of stay (LOS), and improve pulmonary function in non-operative rib fractures [14]. A recent study found EIP to reduce complication rates in an unbalanced mixed cohort of non-operative (80%) and operative (20%) rib fractures compared to standard care with opioids and reported decreased opioid requirements in patients with six or more rib fractures [15].

CryoNB involves the application of pressurized nitrous oxide to the intercostal nerve. The cold results in axonotmesis in which the axon and myelin sheath are injured preventing the sensory nerve from propagating a pain signal. However, the structural elements of the nerve are preserved, facilitating complete regeneration at a rate of 1–2 mm per day thereby restoring normal function over a period of a few months [16,17]. The benefit of this approach centers on its duration of efficacy. CryoNB provides pain control in the acute setting and beyond allowing patients to recover with minimal pain. Previous retrospective studies have shown the concomitant application of CryoNB with SSRF reduces postoperative pain [18] and opioid use [3]. However, a recently published systematic review [19] evaluating CryoNB's pain and opioid-reducing effects in chest wall trauma demonstrated encouraging findings but there was not enough data to make this recommendation. Studies have shown the safety and feasibility of CryoNB when used in conjunction with SSRF [20] as well as lower hospital costs [3].

There is a paucity of research examining outcomes for the concomitant use of CryoNB or EIP adjuncts with SSRF. Furthermore, published studies on pain management for traumatic rib fractures frequently exclude opioid-tolerant patients who represent a growing vulnerable proportion of trauma patients and may comprise the greatest need to employ non-opioid adjuncts with long-term efficacy. Therefore, the aim of this study was to compare the effects of two pain management adjuncts, CryoNB and EIP, applied concurrently with SSRF, on postoperative outcomes including opioid requirements, pain, and pulmonary function in patients sustaining traumatic rib fractures, regardless of preinjury opioid use. Moreover, the effects of these adjuncts on hospital cost, length of stay, and complications were evaluated. We hypothesize that patients treated with CryoNB compared to EIP, would demonstrate favorable results on all outcomes.

Methods

Study population

We performed a retrospective comparative analysis of our Level II trauma centers registry for consecutive patients undergoing SSRF with either intercostal CryoNB or 0.2% ropivacaine EIP from October 2017 to November 2020. Patient demographic, surgical characteristics, and postoperative data were abstracted from the electronic medical record (EMR) including patients age, sex, body mass index (BMI), opioid tolerance (defined as daily use within two weeks prior to presentation and duration of use > 4 weeks, illicit drug use with positive urine or serum toxicology results), the number of ribs fractured, time to operation, admission Glasgow Coma Score (GCS), Injury Severity Score (ISS), rib fracture score, and presence of flail segment. Collected procedural and postoperative data included LOS, number of ribs fixed, opioid consumption, pain scores, pulmonary function, hospital cost, and complications including 30-day readmission and incidence of chest wall neuralgia. Patients younger than 18 years old, pregnant, intubated, sustained a traumatic brain injury, bilateral rib fractures, allergy to local anesthetics, or underwent additional operations were excluded. This study was approved by the institutional review board and the requirement for informed consent was waived due to the retrospective nature of this study.

Pain management

The decision to use either EIP or CryoNB was based on surgeon preference. All patients were placed on standard multimodal pain management protocol per our institution's rib fracture pathway including scheduled and as-needed medications (Table 1). SSFR was approached employing a muscle-sparing technique, fracture preparation for osteosynthesis, direct reduction, and placement of Zimmer Biomet RibFix Blu™ Thoracic Fixation System (Zimmer Biomet: Warsaw, IN, USA) plate(s) and screws, and chest thoracostomy tube placement.

Insertion of the ON-Q elastomeric continuous infusion pump (ON-Q; Lake Forest, CA, USA) occurred at the level of the incision and inferior border of the rib, utilizing a blunt tunneler device (Fig. 1). Under direct visualization, the extrapleural space was then dissected, ensuring the pleural cavity was not entered. The EIP tubing was subsequently inserted into the tunneler to achieve appropriate placement and then removed via a peel-away sheath mechanism. The tubing was secured to the chest wall and the catheter was attached to the pump containing 550 mL of 0.2% ropivacaine at an infusion rate of 5 mL/hour for up to 4 days.

Application of CryoNB was performed at the beginning of the rib fixation procedure under direct visualization via a thoracoscopic approach. AtriCure's 8 mm cryosphere probe (AtriCure, Inc, Mason, OH) was applied to the intercostal nerve (Fig. 2) at the level of each rib fracture, one level above the highest fracture, and one level below the most inferior fracture, taking care to avoid freezing the 3rd intercostal space due to the proximity of the sympathetic trunk or below the 9th

Table 1
Standard analgesic treatment regimen.

Medication	Dose and frequency
Acetaminophen	500 mg po every 6 h
Diazepam	2 mg po every 8 h as needed for muscle spasm
Gabapentin	100–300 mg po three times a day
Ketorolac *or*	15 mg IV every 6 h
Ibuprofen	800 mg po every 8 h
Morphine *or*	1–2 mg IV every 2 h as needed for severe pain (8–10) *
Hydromorphone	0.2- 1 mg IV every 2 h as needed for severe pain (8–10) *
Oxycodone *or*	5 −10 mg po every 4–6 h as needed for moderate pain (5–7) *
Tramadol	50 mg po Q 6 h as needed for mild pain (3–5) *

*Numerical Rating Scale (0 = no pain to 10 = excruciating pain).
IV, intravenous; PO, by mouth.

Fig. 1. Elastomeric infusion pump.

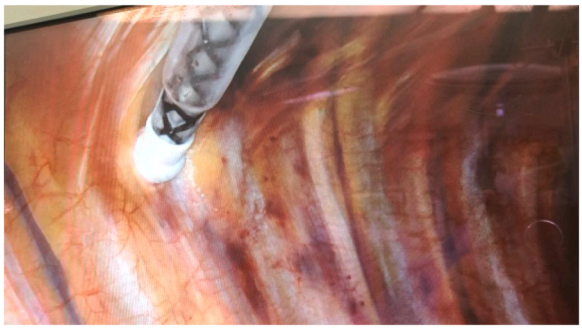

Fig. 2. Application of CryoNB to the intercostal nerve with AtriCure's 8 mm cryosphere probe during SSRF.

intercostal space given the potential for postoperative abdominal wall bulging due to motor innervation to the rectus oblique muscles. As previously described [21], each ablation occurs at a temperature between -50 °C and -70 °C, 4 cm from the base of the spine, for 120 s, and is followed by the active defrost feature which warms the probe to ambient temperature, facilitating removal.

Outcomes

The primary outcome was inpatient opioid consumption measured in milligrams of oral morphine equivalents (OME) and calculated using standardized conversion factors [22]. Total OME consumption was recorded at baseline (24 h prior to surgery) and on postoperative days 1–3.

Secondary outcomes of interest were pain scores measured on an 11-point numeric rating scale (NRS) (0= no pain and 10= worst pain possible) and pulmonary function reflected by incentive spirometry (IS) effort in milliliters (mL). Both outcomes were measured 24 h prior to surgery and again on postoperative days 1–3 by taking the mean value of pain and IS scores recorded in the EMR at the start of each 8-hour shift.

Other outcomes of interest included median length of stay, mean hospital costs, complications resulting in 30-day readmission, and presence of chest wall neuralgia at 1 year following surgery. Median OME prescription at discharge, 30-day opioid refills, and opioid dependence at 1 year following surgery were evaluated for opioid-naive patients only, as opioid-tolerant patients were discharged on their home opioid regimen and we were unable to account for any supplemental illicit drug use.

Statistical analysis

Data analysis was performed in R version 3.2.3 (R Core Team [2013]) using the lme4 package [23] and inferences were done using the lmerTest package [24]. Categorical variables were presented as counts and percentages and analyzed using Pearsons chi-squared or Fishers exact test. Continuous variables were presented as mean (standard deviation [SD]) or median (interquartile range [25th, 75th IQR]) depending on the distribution of the data. Comparisons were made using a two-sample *t*-test or Wilcoxon rank-sum test. A *p*-value of < 0.05 was considered statistically significant.

A linear mixed-effects model was used to estimate the association of CryoNB and EIP on patient outcomes for OME consumption, pain scores, and incentive spirometry effort while controlling for other covariates. This model was selected given repeated measures and concern for violating the assumption of independent error terms. Further, a variance stabilizing transformation of OME values was required given its non-normal distribution. As a result of frequent zero OME values, a logarithmic transformation of OME +1 was performed. Diagnostic plots did not indicate concerning violations of normality assumptions for pain scores or incentive spirometry values, so a transformation was not necessary.

Two random effects were used in our model with the first one denoting a random intercept for each patient accounting for the randomness of the baseline level of the outcome variable for each patient. The second random effect is for the slope associated with each postoperative day. The strength of the relationship between time and the outcome variable may be different for individual patients controlling for other relevant variables. With the addition of these random effects, inference on the fixed effects becomes possible as concerns of independence between errors have been reduced. Random effects were tested using a likelihood ratio test for nested models. Fixed effects were designated as time and group and their interaction [25].

Within the model, we controlled for the following factors: opioid tolerance, sex, age, number of ribs fractured, number of ribs fixed, rib injury severity score (ISS), trauma ISS, presence of flail segment, smoking status, and postoperative length of stay. Additionally, a baseline covariate (first measurement before treatment) was included as a predictor variable to control for differences in the starting values of OME requirements, pain score, and incentive spirometry values. The use of a baseline covariate assists in controlling a large proportion of the variance and increases the statistical power of data analysis [26].

A single observation of an OME value was missing as a result of a patient's postoperative length of stay being only 2 days. The observed variables of this patient were not unusual, so the available measurements remained in the model. Two total pain score measurements were missing, both occurring in a single postoperative measurement for two different patients. Due to the timing of this missingness, parameter estimation was not significantly impacted. A total of 18 incentive spirometry measurements were missing across 12 patients. Fitting a logistic regression model for missingness of the IS variable as a function of observed variables did not result in any statistically significant terms. We were able to model missingness as a function of the observed values suggesting a missing at random (MAR) mechanism. Therefore, the missingness mechanism was ignorable for maximum likelihood estimation. Owing to a lack of published data for postoperative opioid requirements in the setting of concomitant use of CryoNB or EIP with SSRF and inclusive of opioid-tolerant patients, a power calculation was not performed.

Results

Patients' characteristics

From October 5, 2017 to November 10, 2020, a total of 51 patients were identified from our trauma registry who underwent SSRF with

either CryoNB or EIP. After exclusion criteria were applied (multiple operations = 11, significant head injury = 6, transferred > 24 h after injury = 4, SSRF performed electively = 4), 14 CryoNB and 12 EIP patients were included in the study (Fig. 3). Table 2 presents patients demographics and surgical characteristics. There were no significant differences between cohorts. Opioid-tolerant patients comprised 43% (6/14) of the CryoNB cohort and 33% (4/12) of the EIP cohort (p = 0.70). CryoNB versus EIP patients had similar median (range) number of ribs fractured (6 (2–10) versus 7 (4–10), p = 0.18) and repaired (4 (1–5) versus 4 (3–5), p = 0.73) respectively. Mean (SD) days to operative repair was not statistically different for the two cohorts: CryoNB: 3.1 (1.4) versus EIP: 3.6 (2.5), p = 0.67. During the study, no patients were lost to follow-up.

Outcomes

The primary outcome, inpatient opioid consumption, is shown in Fig 4 and Table 3. CryoNB resulted in a 25% (95% CI: 9,70, p = 0.01) reduction in opioid consumption following surgery. With respect to opioid requirements at discharge, opioid-naive CryoNB patients were prescribed fewer median opioids by 133.7 mg (p = 0.03), and 38% (3/8) of these patients compared to 13% (1/8) EIP patients did not require a discharge opioid prescription (p = 0.56) (Table 4). One patient in each cohort required an opioid prescription refill within 30 days of discharge.

Secondary outcomes for change in pain scores from the preoperative to postoperative time period resulted in CryoNB patients experiencing a significant reduction of 22% (95% CI: 7, 75, p = 0.02). Pulmonary function showed no difference between groups (p = 0.71) (Table 3). The same insignificant trend was seen with total hospital LOS. CryoNB patients were discharged approximately 2 days sooner (9 (7–10) versus 11 (9–15.5), p = 0.09). However, median postoperative LOS was significantly shorter in the CryoNB cohort (4 (4–5) versus 6 (5–9.5), p = 0.04). As seen in Table 5, mean hospital costs were also lower for patients treated with CryoNB by $41,274 ($p$ = 0.07) and EIP patients had overall greater costs regardless of opioid status. Complications for all-cause readmission within 30 days following the index operation demonstrated no difference between groups, with 1 opioid-tolerant patient from each cohort readmitted for pain control. Two EIP patients had persistent chest wall pain at 1 year following surgery and 1 EIP opioid-naive patient became opioid dependent at 1 year following surgery.

Table 2
Patient demographic characteristics.

	CryoNB (n = 14)	EIP (n = 12)	P value
Age (years), mean (SD)	54 (12.3)	52 (20.5)	0.73
Male, n (%)	11 (78)	7 (58)	0.40
BMI (kg/m2), mean (SD)	30.8 (7.2)	33.1 (8.8)	0.52
Tobacco use, n (%)	6 (43)	3 (25)	0.43
Opioid tolerant, n (%)	6 (43)	4 (33)	0.70
Number of ribs fractured, median (range)	6 (2–10)	7 (4–10)	0.18
Number ribs fixated, median (range)	4 (1–5)	4 (3–5)	0.73
Time to surgery (days), mean (SD)	3.1 (1.4)	3.6 (2.5)	0.67
Admission GCS, median (IQR)	15 (6–15)	15 (14–15)	>0.99
ISS, mean (SD)	15.9 (8.7)	21.4 (10.3)	0.23
Rib fracture score, median (range)	9 (4–16)	9 (9–16)	0.81
Flail segment, n (%)	7 (50)	5 (42)	0.71

CryoNB = cryo nerve block, EIP = elastomeric infusion pump, GCS = glosgow coma score, IQR = interquartile range, ISS = injury severity score, SD = standard deviation.

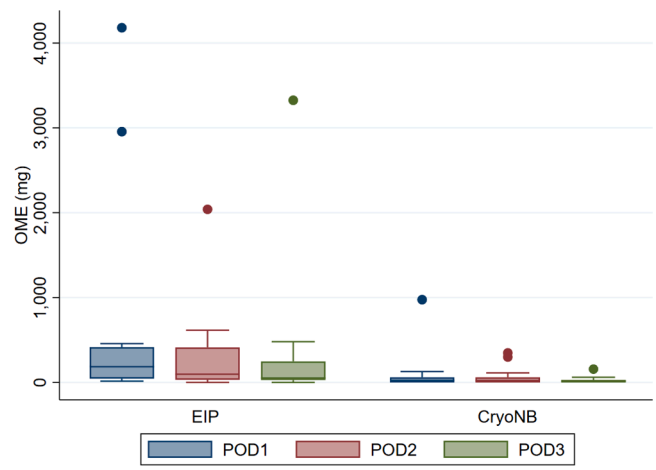

Fig. 4. Postoperative trend of OME (mg) requirements by cohort.

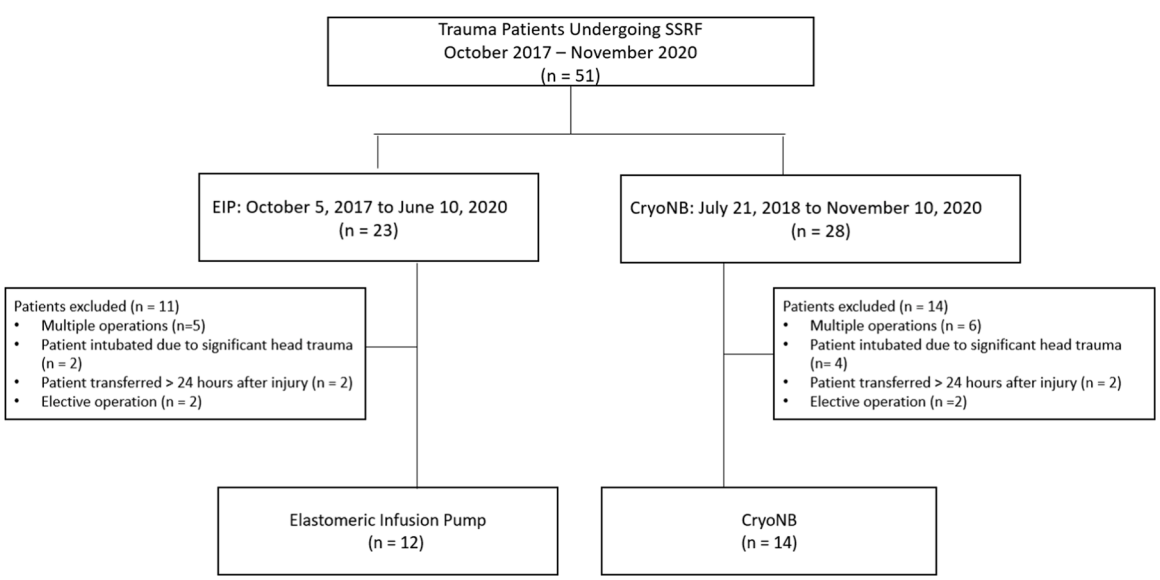

Fig. 3. Flow diagram describing patient population and exclusions.

Table 3
Outcomes from linear mixed effects model.

Fixed effect	Point estimate	Standard error	Estimated *df*	*t* -value	95% CI	*P* value
OME	-1.373	0.52	68.8	-2.638	-2.411,-0.335	0.01
Pain	-1.5061	0.61025	73.138	-2.468	-2.722, -0.290	0.02
IS	-48.843	131.836	58.42	-0.37	-312.74, 215.055	0.71

IS = incentive spirometer, OME = = oral morphine equivalent.

Table 4
Opioid naive specific outcomes.

	CryoNB (*n* = 8)	EIP (*n* = 8)	*P* value
Discharge opioid prescription (mg), median [IQR]	41.25 [37.5,45]	175 [150,200]	0.03
No discharge opioid prescription, n (%)	3 (38)	1 (13)	0.56
Opioid prescription refill 30 days after discharge, n (%)	1 (13)	1 (13)	>0.99
Patients on opioids 1-year after surgery, n (%)	0	1 (13)	>0.99

CryoNB = cryo nerve block, EIP = elastomeric infusion pump, IQR = interquartile range.

Table 5
Secondary outcomes.

	CryoNB (*n* = 14)	EIP (*n* = 12)	*P* value
Total Hospital LOS (days), median (IQR)	9 (7–10)	11 (9–15.5)	0.09
Postoperative LOS (days), median (IQR)	4 (4–5)	6 (5 - 9.5)	0.04
Postoperative Chest wall neuralgia at 1 year, n (%)	0 (0)	2 (17)	0.11
30-Day All-Cause Readmission, n (%)	1 (7)	1 (8)	>0.99
Total Hospital Costs (all patients), mean (SD)	90,224 (34,633)	131,498 (73,072)	0.07
Total Hospital Costs (opioid naïve), mean (SD)	(*n* = 8) 91,080 (37,440)	(*n* = 8) 125,161 (69,776)	0.24
Total Hospital Costs (opioid tolerant), mean (SD)	(*n* = 6) 89,083 (33,963)	(*n* = 4)144,170 (88,868)	0.19

CryoNB = cryo nerve block, EIP = elastomeric infusion pump, GCS = glasgow coma score, IQR = interquartile range, ISS = injury severity score, LOS = length of stay, OR = operating room, SD = standard deviation.

Discussion

The results of our study suggest that CryoNB may have a clinical advantage over EIP, including a statistically significant decrease in hospital OME consumption by 25%, pain scores by 22%, fewer discharge opioids, and shorter postoperative LOS following SSRF. Pulmonary function, hospital costs, and complications were not significantly different between cohorts. Our findings are similar to previous studies for the concomitant use of CryoNB with SSRF [3,18].

A systematic review by Cha et al. [19], examined the efficacy of intercostal CryoNB for chest wall pain sustained from surgery or trauma. While a majority of the included studies demonstrated decreased inpatient opioid use with CryoNB compared with conventional pain management, the evidence was of low quality, and therefore no recommendation was made. Nonetheless, the authors concluded a favorable risk-benefit profile and potential to provide long-term pain control in rib fracture patients who failed medical management. Our study results are consistent with a recently published retrospective study of 44 patients receiving CryoNB in conjunction with SSRF versus SSRF alone and showed a reduction of opioid requirements by 25.1 mg [3]. However, a pilot study by Choi et al. [20], demonstrated no statistically significant difference in pain, opioid requirements, or LOS in 12 patients treated with CryoNB and SSRF compared to 8 patients receiving standard of care and SSRF but found CryoNB to be safe and feasible.

EIP, similar to CryoNB, doesn't require ongoing monitoring and eliminates the risks associated with other modalities, but is only effective for as long as it remains in place. Despite this, previous studies showed that EIP promotes earlier mobilization [13] and reduced pain following thoracotomy compared to thoracic epidurals [11], yet others have found no difference [10]. However, there remains a risk of systemic toxicity [12,13] and one study found that 16% of patients developed a wound infection following a thoracotomy with EIP [12]. No EIP patients in our study experienced these complications. EIP has been shown to reduce pain and opioid requirements for the treatment of non-operative rib fractures [14], but only one retrospective study included a small proportion of patients treated with SSRF [15]. The authors found EIP to significantly lower opioid requirements for patients with 6 or more rib fractures compared to standard care [15]. However, the authors did not specifically report opioid use or pain scores for the subgroup of patients undergoing SSRF.

The clinical indications for cryotherapy have been established since the 1950s for the limited treatment of peripheral nerves. Its application is increasingly being adopted within thoracic surgery as a viable option for pain management. Despite a growing body of published evidence concerning the pain-lowering and opioid-sparing benefits of CryoNB within thoracic surgery, there remain concerns about its role in creating neuropathic pain. CryoNB preserves the critical structures (endoneurium, perineurium, epineurium) necessary for nerve regeneration preventing neuroma formation [16,27]. The refinement of technology established optimal freezing duration and temperature, producing greater consistency in outcomes compared to historical models and previously published data referencing this risk [19]. Injury to the thoracic cage and subsequent surgical intervention can both lead to intercostal nerve damage and resultant neuropathic pain from possible disorganized nerve regeneration, making it challenging to discern the true origins of nerve injury [18,28]. Current literature shows that patients treated with CryoNB and report postoperative neuralgia resolve by 6 months [18]. We found 1 opioid-tolerant patient from each cohort required readmission within 30 days following surgery for management of chest wall pain, with both patients reporting resolution by 3 months. Two patients treated with EIP reported neuralgia 1 year following surgery. The CryoNB cohort also had 2 patients reporting neuralgia at 3 months following surgery but these symptoms resolved by 6 months. However, one of these patients had persistent numbness 1 year after surgery but was not bothersome.

The proposed benefit of CryoNB is its duration of efficacy, beginning at the time of surgery and lasting up to 3 months following application. This extended duration facilitates effective pain control over an expected long period of healing making it more advantageous than short-lived analgesic effects found in conventional regional techniques. Fagundes et al. [29] showed that it takes 42 days for patients to reach 80% recovery equivalent to mild pain levels following a thoracotomy and 25 days to reach the same threshold with less invasive video thoracoscopy. Providing effective pain control during this transitional period, from hospital to recovery is critically important, as poorly controlled transitional pain increases the risk of developing chronic pain 3 months after surgery [30,31]. In our study, 1 opioid-naive patient treated with EIP remained on opioids for persistent chest wall pain 1 year following surgery while no opioid-naive CryoNB patients required chronic opioids. Furthermore, CryoNB opioid-naive patients were discharged home with significantly fewer opioids compared to the EIP cohort. This may be due to individual prescribing habits or increased pain at discharge following EIP removal. Notably, 37.5% of opioid-naive patients treated with CryoNB did not require an opioid prescription at discharge compared to 12.5% of EIP patients.

Consistent with prior studies [21,32], pulmonary function was equivocal between groups and this may be related to the small sample size or patients' inability to create a tight seal around the mouthpiece. Similar to previously published results [3], we found a statistically significant shorter postoperative LOS in CryoNB patients by approximately 2 days. While not statistically significant, CryoNB patients had lower total hospital costs by $41,274. The financial implications of lower hospital charges offset the approximate $2500 cost of CryoNB. Although charges for an EIP were lower ($542), cost savings may be realized by fewer opioid requirements (medication costs), decreased rates of readmission for complications, and long-term benefits gained from fewer circulating opioids in the community as well as the potential to mitigate any increase in opioid-tolerant patients baseline opioid requirements. Not surprisingly, in our study, opioid-tolerant patient's hospital charges were $17,012 greater than opioid-naive patients and opioid-tolerant patients treated with EIP had higher hospital costs of $55,086 and a longer LOS of 1.4 days.

The Opioid epidemic in North America has been at the forefront of public health discussions since the late 1990s [33]. Its origins are attributed to the rise of opioid prescriptions and subsequent increased consumption. Opioid-sparing and non-opioid alternatives to control pain in opioid-naive and tolerant patients are necessary. Currently published research centers on opioid reduction strategies to prevent the development of opioid addiction but rarely are opioid-tolerant patients included. Likely, this may be attributed to challenges in analyzing this patient population given the complex and subjective nature of one's perception of pain. Furthermore, the literature demonstrates that patients involved in traumatic accidents are frequently under the influence of opioids and predisposed to sustaining injuries requiring treatment with opioids [7]. Therefore, future studies centered on surgical adjuncts to achieve extended duration of opioid-sparing pain control will improve how we manage pain in opioid-naive and tolerant patients alike.

Limitations

There are several limitations to our study. The nature of a retrospective study limits our ability to make definitive estimates for the effectiveness of the treatment interventions and may account for an increase in type II errors. Despite the increased adoption of SSRF, the incidence of SSRF in trauma centers remains low [34] and is apparent in our small sample size. This coupled with a notable absence of published studies regarding opioid requirements following SSRF, inclusive of opioid-tolerant patients, impacted our ability to perform a power analysis. However, the statistical methods employed herein are robust in the ability to account for each individual patient change over time. Although this approach may help account for unmeasured cofounders [18,35], we remain limited in our ability to generalize our findings. We were unable to include a control group of SSRF patients treated with conventional analgesia but doing so would demonstrate the magnitude of each adjunct's pain and opioid-reducing effects beyond SSRF alone. Additionally, the integrity of our data is dependent upon the accuracy of the medical record. Further, if we had access to hospital charges on a daily basis rather than as an aggregate, we may have observed a significant difference in postoperative charges between cohorts. Finally, patients classified as opioid tolerant included those using illicit drugs and understanding the impact this may have on opioid consumption is unclear. Outcomes from this study must be considered with these limitations in mind and should be interpreted as hypothesis-generating. Despite these limitations, this study represents the first evaluation comparing CryoNB with EIP in conjunction with SSRF.

Conclusion

The concomitant use of CryoNB during SSRF resulted in lower opioid requirements and pain scores compared to EIP with SSRF. These findings are inclusive of opioid-tolerant patients who represent a growing proportion of the trauma population poised to significantly benefit from opioid-sparing adjuncts. Furthermore, our findings suggest that CryoNB may also reduce patients' hospital length of stay and overall costs. CryoNB's favorable attributes include its ease of application, safety profile, and extended duration of pain control. Additional research with large prospective randomized trials, inclusive of opioid-tolerant patients and a medically managed control group, is needed in order to elucidate these early findings.

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Declaration of Competing Interest

L.O'Connor and C.Quinn serve as educational faculty to AtriCure®. B.Houseman and C.Quinn serve as educational faculty to Zimmer-Biomet. T. Cook has nothing to declare.

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